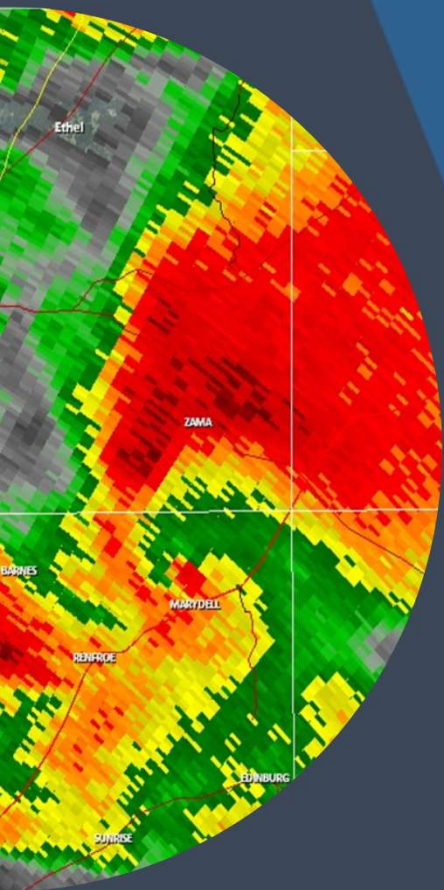




Identifying User Needs for the Future of Weather Radar: A Town Hall Discussion on Radar Next

National Weather Service

15 January 2025





Agenda



1215-1230

- Overview of Radar Next Program and Activities

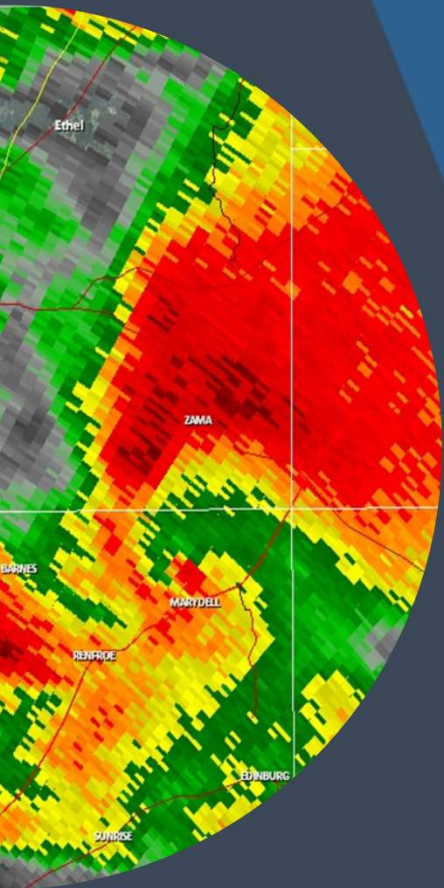
1230-1245

- User Panel Introduction and Remarks

1245-1315

- Panel Discussion and Audience Q&A – Discussion of User Needs for Radar Next





Identifying User Needs for the Future of Weather Radar: A Town Hall Discussion on Radar Next

Frank W. Gallagher III, Jessica Schultz, Mat Grow,
Ajay Mehta, and Terrance Clark
NOAA/NWS/OBS



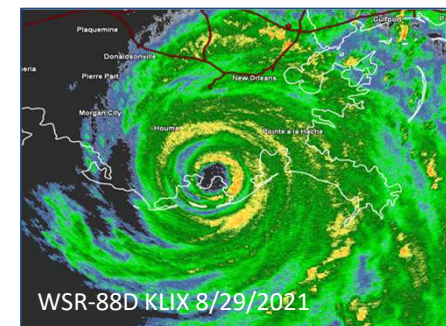
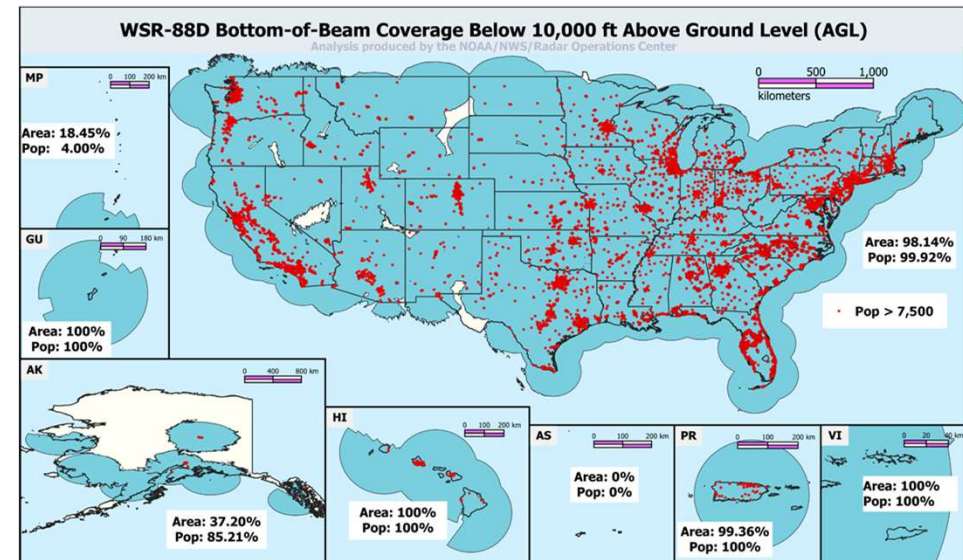
National Weather Service

15 January 2025

NEXRAD Program: Current Operational Weather Radar



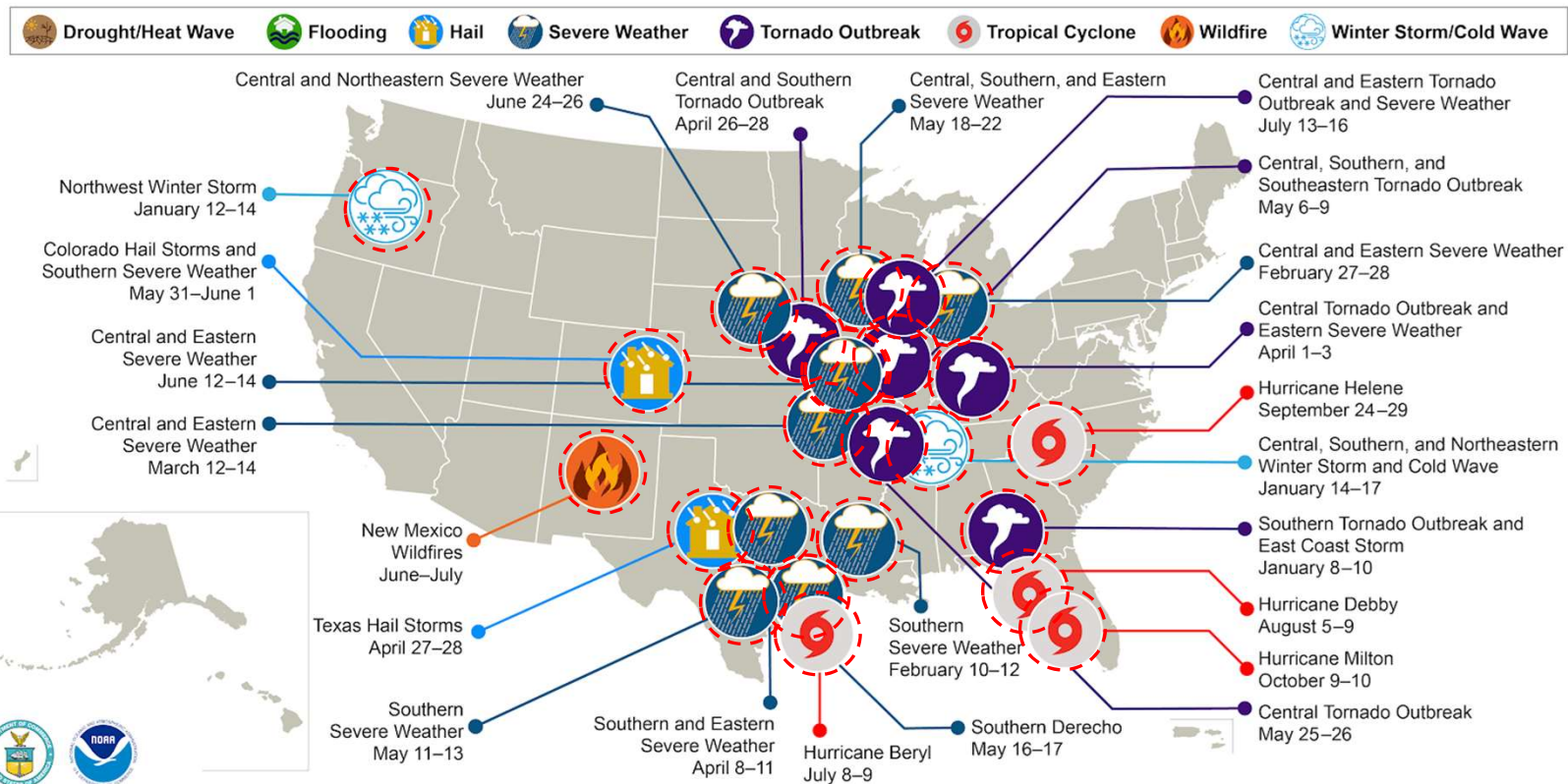
- Network of 158 S-band Doppler weather radars jointly operated by NWS, USAF, FAA
- Provides mosaic coverage of most of the Continental United States
- Most impactful NOAA observational asset supporting severe weather, flash floods, and air traffic management
- Installed in the 1990s with a 20-year expected design life
- Completed Dual Polarization upgrade 2007 - 2013 and Service Life Extension Program 2015 - 2024



Why It Matters



U.S. 2024 Billion-Dollar Weather and Climate Disasters



This map denotes the approximate location for each of the 24 separate billion-dollar weather and climate disasters that impacted the United States through October 2024.



National Weather Service
Office of Observations

Threats to NEXRAD Availability



Damaged Components



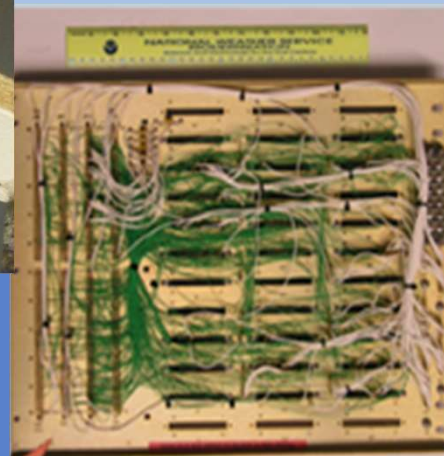
Metal Fatigue



Environmental & Wiring Issues



Obsolete Parts

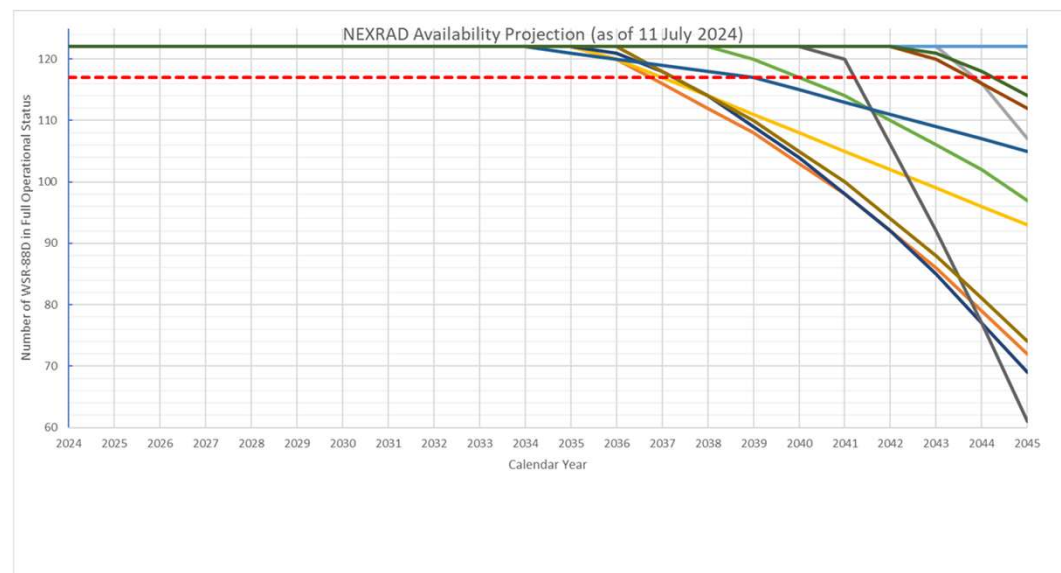


Threats to NEXRAD Availability



NEXRAD has exceeded its original design life

- Aging infrastructure leading to reduced reliability, availability, and maintainability
- Obsolescence / Unavailability of commercial parts
- Third-party supply chain disruptions
- Inflationary pressures (e.g., NEXRAD optical position encoders increased in price from ~\$4K to \$44K)



B. Foltz, A Study on the Future Availability of NEXRAD - Study Approach [abstract] In: American Meteorological Society 105th Annual Meeting; 13 Jan 2025, New Orleans, LA, Abstract 1.1.

Availability will fall below requirement by mid 2030s



National Weather Service
Office of Observations

The New NWS: Partner Focused Operating Model



To best meet our mission,

alongside our partners,
we must be...

- 📍 located where those decision-makers need us.
- 📍 available when they need us.
- 📍 deliver products and services in the manner in which they are needed.

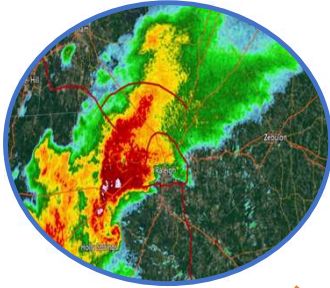
Those needs are constantly changing,
and we must too...

NWS Mission

To provide weather, water and climate data, forecasts, warnings, and impact-based decision support services for the protection of life and property and enhancement of the national economy.



Radar Next - Program Drivers

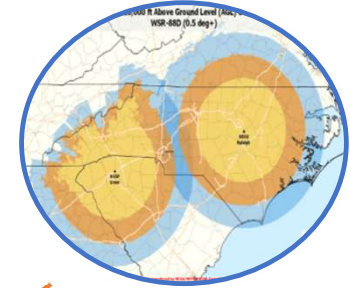


Data Continuity

- Prevent lapse in Availability
- Maintain high quality dual-polarization radar data

Improve Data Performance

- Integrate new science
- Data quality
- Improve update rate
- Spectrum congestion
- Sidelobes / Ground Clutter / Wind Farms



Improve Coverage

- Improve low-level coverage
- Greater geographic coverage
- Inclusion of traditionally underserved communities



Radar Next - Paradigm Shift



Integration of data from multiple sources into a single grid and display

Improved geographical coverage

Improved Data Accessibility



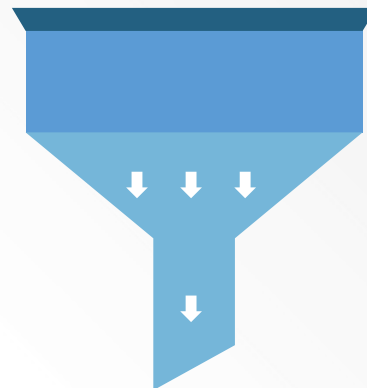
Improved spatial and temporal resolution



Assimilation from hybrid radar networks into numerical forecast models



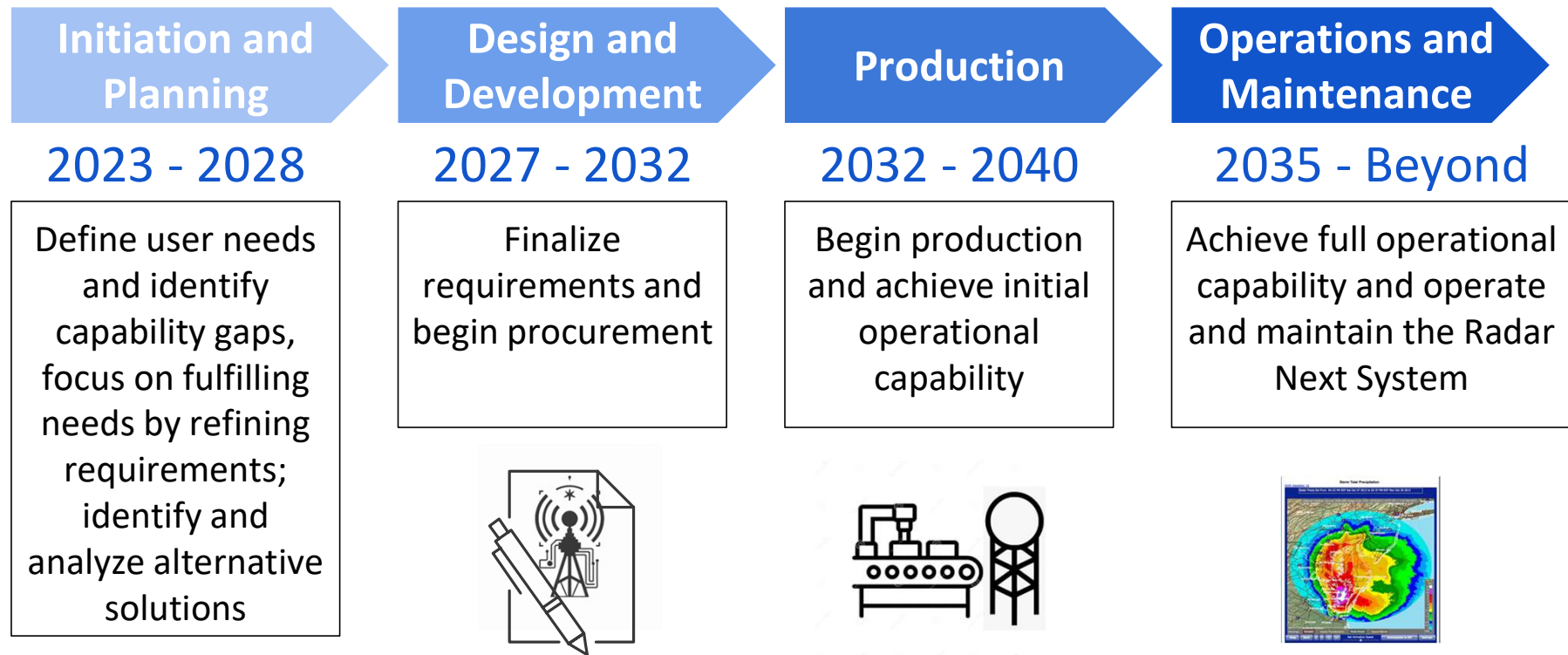
Improved coverage at lower altitudes



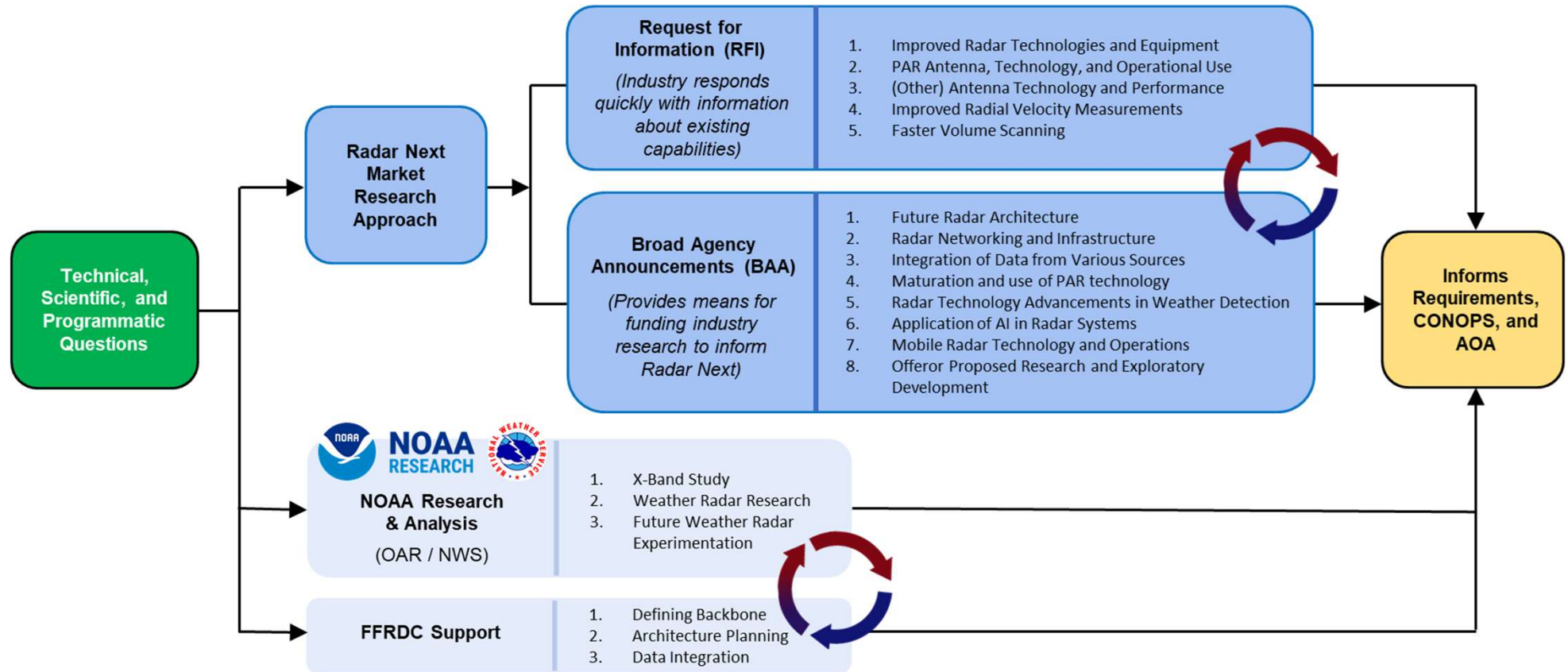
Improved User Experience



Radar Next Program's Timeline



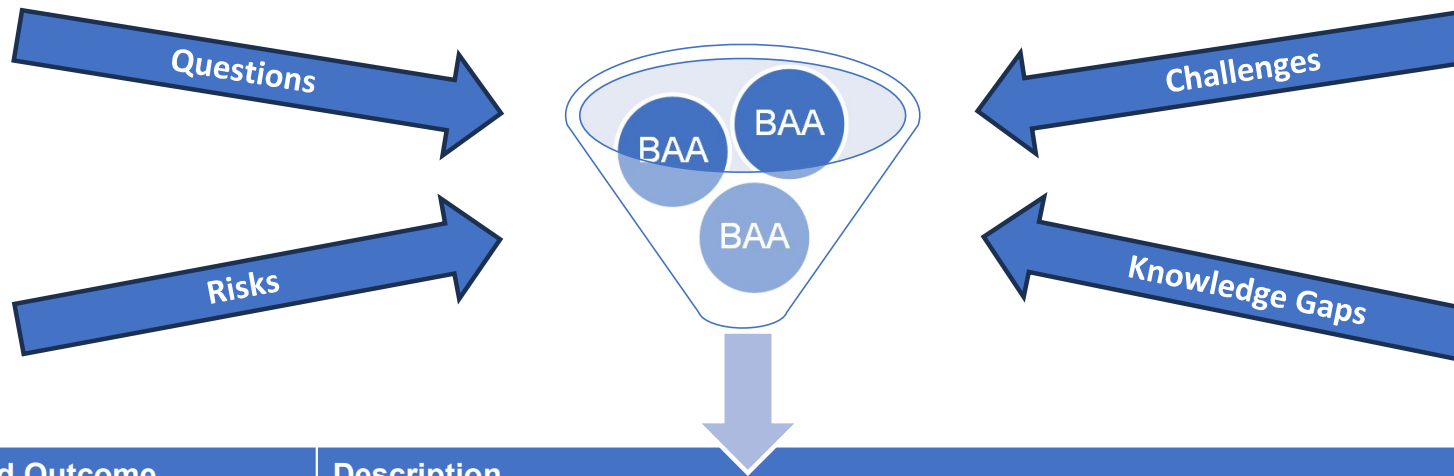
Ongoing Radar Next Engagements



Subject to FAR 52.232-18 Availability of Funds



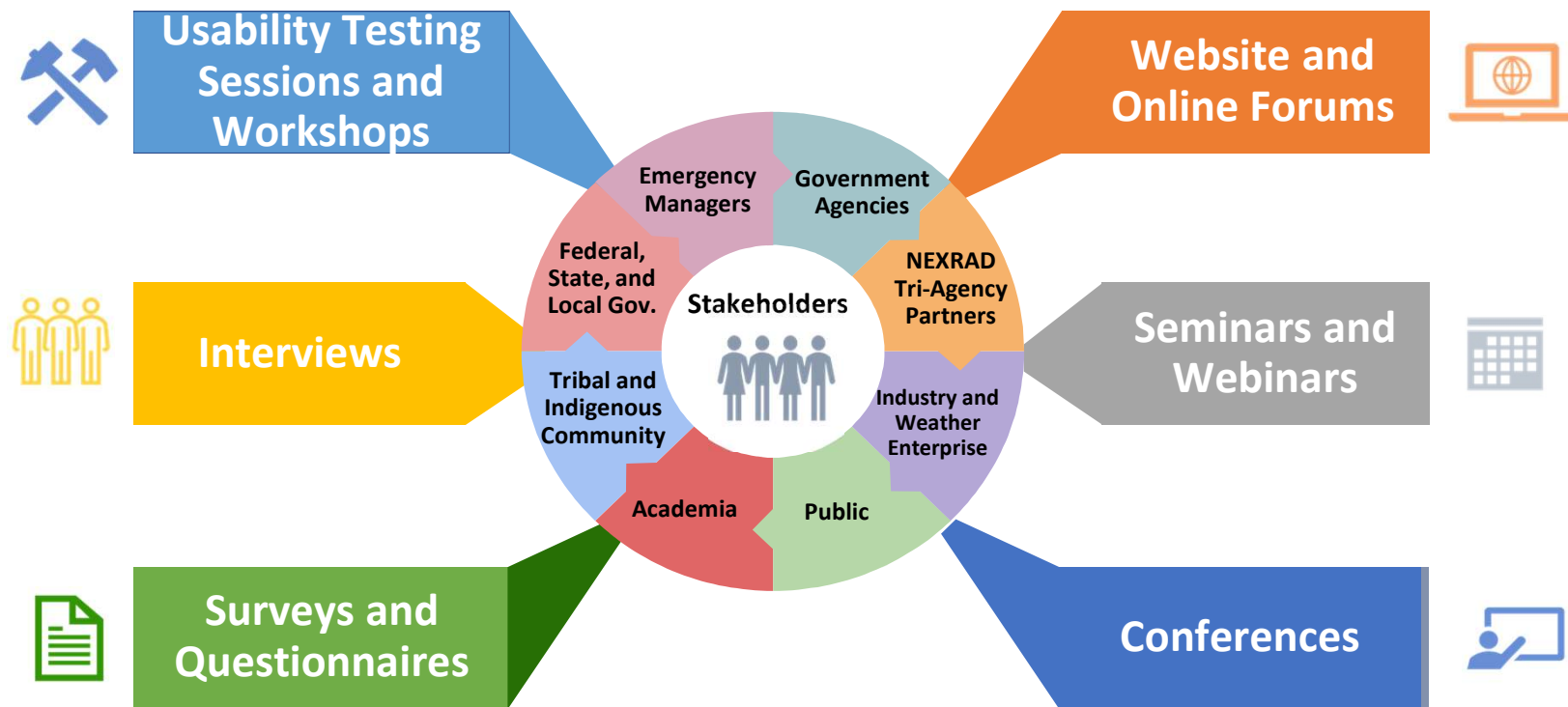
Broad Agency Announcements (BAAs)



Expected Outcome	Description
Broaden Scope of Solutions	By soliciting ideas from a wide audience, BAAs introduce <u>novel technologies and methodologies</u> and <u>diverse and innovative solutions</u> that might not be considered otherwise.
Cutting-edge Research	BAAs often attract proposals that incorporate the <u>latest research and technological advancements</u> , enriching the AoA process.
Foster Collaboration	BAAs foster collaboration between agencies and the research community, leading to a <u>rich exchange of ideas and expertise</u> .
Flexibility and Adaptability	The open-ended nature of BAAs allows for <u>flexibility in exploring unconventional solutions</u> .

Collection of User Needs and Assessments

Radar Next will conduct many user engagement activities over the next several years.



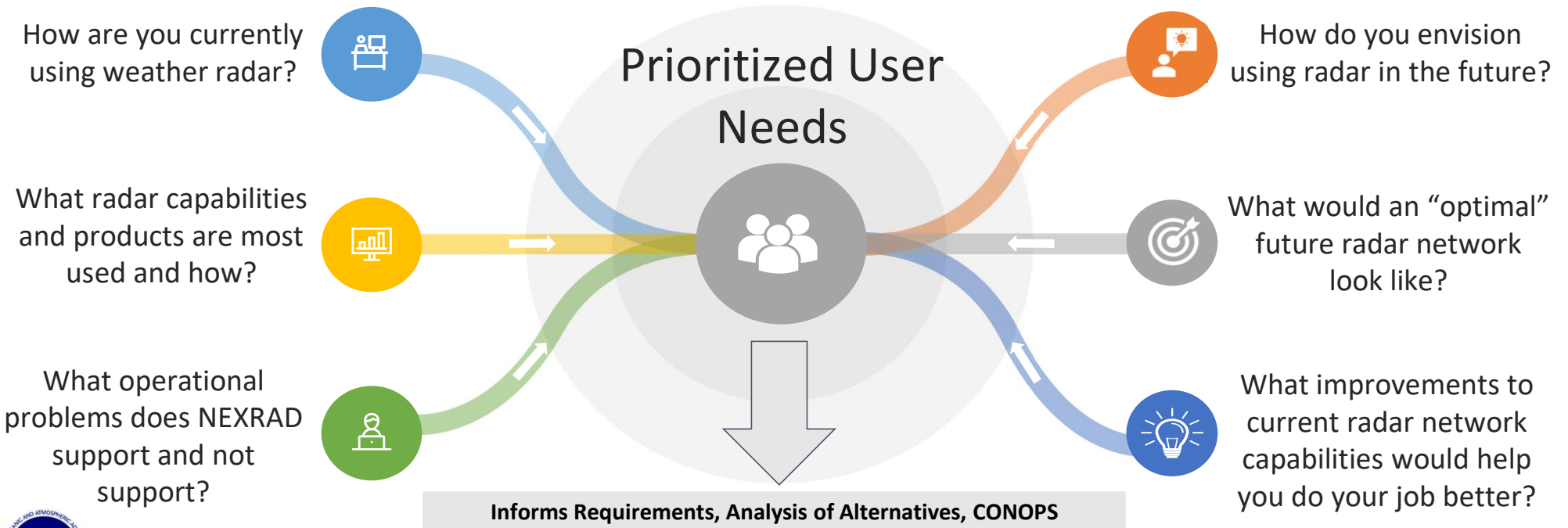


Start with User Needs: Collection Approach

Radar Next is collecting the full scope of validated, technically realizable, Doppler weather radar user needs that are implementation agnostic and without regard to budget

Current Needs

Future Needs





Radar Next: What to Expect Next



Develop Program Requirements and Alternative Solutions

- Collect the full scope of validated, technically realizable, Doppler weather radar user needs
- Establish the framework for developing and evaluating the cost-benefit of possible solutions

Investigate New Concepts of Operations

- Change the paradigm of radar data collection to be more flexible and agile
- Explore new command/control and maintenance philosophies

Build Upon and Improve Weather Radar Capabilities

- Incorporate feedback from stakeholders and end users
- Once in a generational opportunity to influence the next generation of weather radar capabilities

Contact Us: radar.next@noaa.gov





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Introductions



- **Mike Fowle**, Science and Operations Officer, NWS Des Moines, Iowa
- **Randall Bass**, Owner and President of Bass Weather Services
- **Ashley Morris**, Certified Emergency Manager, Maryland
- **Dr. Frank W. Gallagher III**, Meteorologist and Systems Engineer, Radar Next

